HastaDAQ

Net labels connect everything: same name = same net. +3V3 / GND are plain labels, not power symbols.

Pimoroni Pico Plus 2 (+ RM2 on SP/CE): same 40-pin header as Pico 2 W. I2C0 = GP4/GP5 (header pins 6/7), hard-wired to the IMU Qwiic-breakout header and both ADS122C04 boards by header pins. Crossed-out pins are intentionally unused.

J1
IMU via Qwiic breakout (The Pi Hut 106676)

GND 1
+3V3 2
SDA 3
SCL 4
GND
3V3
SDA
SCL

ADS1 supply decoupling (place at ADC)

+3V3
C1
10uF
GND
+3V3
C2
100nF
GND

J8
SBUS servo header 3x6 (6 plugs: GND, V, SIG)

GND 1 S1_GND
VSERVO 2 S1_V
SBUS 3 S1_SIG
GND 4 S2_GND
VSERVO 5 S2_V
SBUS 6 S2_SIG
GND 7 S3_GND
VSERVO 8 S3_V
SBUS 9 S3_SIG
GND 10 S4_GND
VSERVO 11 S4_V
SBUS 12 S4_SIG
GND 13 S5_GND
VSERVO 14 S5_V
SBUS 15 S5_SIG
GND 16 S6_GND
VSERVO 17 S6_V
SBUS 18 S6_SIG

TEST POINTS (2.0 mm through-hole pads): two GND pads for a scope clip (near the supply and near the ADCs). Rails and signals are probed at component leads and header pins (fuse clips, C5/C6/C7, D1, Pico pins 36/39).

GND
TP1
TP
GND
TP2
TP

PWR_FLAG: marks nets supplied through the supply connector (+12V_IN, GND), the fuse (+12V) and the Schottky (+5V_VSYS)

#FLG1
PWR_FLAG
+12V_IN
#FLG2
PWR_FLAG
+12V
#FLG3
PWR_FLAG
GND
#FLG4
PWR_FLAG
+5V_VSYS

U1
Pimoroni Pico Plus 2 (header pins)

GP0_UART0-TX 1
GP1_UART0-RX 2
GND 3 GND
GP2 4
GP3 5
GP4_I2C0-SDA 6 SDA
GP5_I2C0-SCL 7 SCL
GND 8 GND
GP6 9
GP7 10 ENC4_CS
GP8 11 ENC3_CS
GP9 12 ENC2_CS
GND 13 GND
GP10_SPI1-SCK 14 SPI_SCK
GP11_SPI1-TX 15 SPI_MOSI
GP12_SPI1-RX 16 SPI_MISO
GP13_SPI1-CSn 17 ENC1_CS
GND 18 GND
GP14 19 TRIG_IN_GPIO
GP15 20 TRIG_OUT_GPIO
VBUS 40
VSYS 39 +5V_VSYS
GND 38 GND
3V3_EN 37
3V3(OUT) 36 +3V3
ADC_VREF 35
GP28 34
AGND 33 GND
GP27 32
GP26 31
RUN 30
GP22 29
GND 28 GND
GP21_UART1-RX 27 SBUS
GP20_UART1-TX 26 SBUS
GP19 25
GP18 24
GND 23 GND
GP17 22
GP16 21 SOFTSTOP_IN

U2
Adafruit ADS122C04 #1 (I2C 0x40)

+3V3 1 VIN
GND 2 GND
SDA 3 SDA
SCL 4 SCL
5 RST
6 DRDY
A0 7 ADS1_SIG
A1 8 ADS_REF
A2 9
A3 10
Ref+ 11
Ref- 12

J2
HB1 half-bridge #1 (on beam)

+3V3 1 V
ADS1_SIG 2 SIG
GND 3 GND

Shared completion resistors (fixed) -> A1 of BOTH ADCs

+3V3
R1
350R fixed
ADS_REF
R2
350R fixed
GND

ADS2 supply decoupling (place at ADC)

+3V3
C3
10uF
GND
+3V3
C4
100nF
GND

J4
ENC1 SPI port: MT6835 (pitch) / Orbis

GND 1 GND
ENC1_CS 2 CS
SPI_SCK 3 SCK
SPI_MOSI 4 MOSI
SPI_MISO 5 MISO
+3V3 6 3V3
+5V 7 5V
GND 8 GND

U3
Adafruit ADS122C04 #2 (addr jumper bridged)

+3V3 1 VIN
GND 2 GND
SDA 3 SDA
SCL 4 SCL
5 RST
6 DRDY
A0 7 ADS2_SIG
A1 8 ADS_REF
A2 9
A3 10
Ref+ 11
Ref- 12

J3
HB2 half-bridge #2 (on beam)

+3V3 1 V
ADS2_SIG 2 SIG
GND 3 GND

J9
Main supply cable (2x4): VSUPPLY, VGND, TRIG, SOFTSTOP

+12V_IN 1 VSUPPLY
GND 2 VGND
TRIG_IN 3 TRIG_IN
SOFTSTOP_CABLE 4 SOFTSTOP+
GND 5 SOFTSTOP-
TRIG_OUT 6 TRIG_OUT
GND 7 VGND
+12V_IN 8 VSUPPLY

J5
ENC2 SPI port: MT6835 / Orbis

GND 1 GND
ENC2_CS 2 CS
SPI_SCK 3 SCK
SPI_MOSI 4 MOSI
SPI_MISO 5 MISO
+3V3 6 3V3
+5V 7 5V
GND 8 GND

TRIGGER IN: any 3.3 V .. 24 V level, opto input

TRIG_IN
R9
10k pull-down
GND
TRIG_IN
R10
1k 1W
TRIG_IN_LED
U6
PC817 opto (TRIG_IN)
TRIG_IN_LED 1 A
GND 2 K
4 C
3 E
TRIG_IN_GPIO
GND
+3V3
R11
10k pull-up
TRIG_IN_GPIO
D3
1N4148 (LED reverse protect)
GND A
K TRIG_IN_LED

J6
ENC3 SPI port: MT6835 / Orbis

GND 1 GND
ENC3_CS 2 CS
SPI_SCK 3 SCK
SPI_MOSI 4 MOSI
SPI_MISO 5 MISO
+3V3 6 3V3
+5V 7 5V
GND 8 GND

CS pull-ups (10k)

+3V3
R3
10k
ENC1_CS
+3V3
R4
10k
ENC2_CS
+3V3
R5
10k
ENC3_CS
+3V3
R6
10k
ENC4_CS

SOFTSTOP_CABLE
R7
1k (soft-stop series)
SOFTSTOP_IN
+3V3
R8
10k (soft-stop pull-up)
SOFTSTOP_IN

J7
ENC4 SPI port: generic SPI device

GND 1 GND
ENC4_CS 2 CS
SPI_SCK 3 SCK
SPI_MOSI 4 MOSI
SPI_MISO 5 MISO
+3V3 6 3V3
+5V 7 5V
GND 8 GND

D2
5.1V zener (BZX55C5V1) soft-stop clamp
GND A
K SOFTSTOP_IN

TRIGGER OUT: open-collector NPN, pull-up rail chosen by jumper

TRIG_OUT_GPIO
R12
2.2k (base)
TRIG_OUT_BASE
TRIG_OUT_BASE
R13
10k (base-emitter)
GND
Q1
2N3904 NPN (TO-92, pin 1 E, 2 B, 3 C)
TRIG_OUT_BASE 2 B
3 C TRIG_OUT
1 E GND
TRIG_PU
R14
1.5k
TRIG_OUT
J10
TRIG_OUT pull-up sel.
+3V3 1
TRIG_PU 2
+5V 3
TRIG_PU 4
+12V 5
TRIG_PU 6
3V3
PU
5V
PU
12V
PU

U4
Pololu D42V55F8 8.4V 5A buck (servo rail)

VSERVO 1 VOUT
GND 3 GND
GND 5 GND
+12V 7 VIN
9 VRP
11 PG
VOUT 2 VSERVO
GND 4 GND
GND 6 GND
VIN 8 +12V
VRP 10
EN 12

+12V_IN
F1
Fuse T3.15A 5x20mm time-delay
+12V
+12V
C5
100uF input bulk (>=25V)
GND
VSERVO
C6
470–1000uF servo bulk (>=16V)
GND

U5
SparkFun BabyBuck 5V (AP63357) (Pico VSYS, Orbis)

+12V 3 VIN
GND 2 GND
4 EN
VOUT_5V 1 +5V

D1
Schottky (e.g. 1N5822, 3A)
+5V A
K +5V_VSYS

+5V
C7
100uF 5V bulk (>=10V)
GND

Jumper a row (3V3 / 5V / 12V) to pull TRIG_OUT up to that rail: remove it if the receiver has its own pull-up.
Pull-up current: 3.3 V ~2 mA, 5 V ~3 mA, 12 V ~7 mA (enough for a 6.5 mA opto input such as MPS4200).
24 V receivers: no jumper, add an external 24 V pull-up in the harness. 3V3 for eSync 2 high-Z ports (3.3 V max).

NOTES
1. ONE common GND (Pico, sensors, servo PSU). SBUS needs a shared ground reference.
   RM2 WiFi breakout (off-board, cabled to the Pico Plus 2 SP/CE connector): uses GP32–GP36 only, no overlap with I2C (GP4/5) or any other signal.
2. ADS2: bridge one address jumper on its back (A0/A1); resulting address unconfirmed -- firmware falls back to an I2C scan at boot. I2C pull-ups: rely on the Adafruit breakouts' on-board pull-ups.
3. Each half-bridge module (2 gauges) plugs into a 3-pin header: V, SIG (mid-point), GND. SIG -> A0 of its ADC, direct (no input filter); ONE shared pair of fixed 350R completion resistors: its mid-point (ADS_REF) goes to A1 of BOTH ADCs. Excitation is the ADC's own VIN, so VREF_SUPPLY is ratiometric -- no Ref+/Ref- wiring.
4. SBUS: GP20 (TX) and GP21 (RX) on one net (UART inverted in firmware). One 3x6 servo header (6 plugs of GND, V, SIG: footprint HastaDAQ:PinHeader_3x06): V = VSERVO (8.4 V), SIG = SBUS (all six in parallel), GND.
5. POWER: 12 V in -> fuse -> +12V rail -> (a) Pololu D42V55F8 = VSERVO 8.4 V (servos), (b) SparkFun BabyBuck 5V = +5V (Orbis, Schottky -> Pico VSYS). VSYS max is 5.5 V. USB (VBUS) is left for programming; with USB only, +5V is not live. BabyBuck EN is left open (its own divider enables it). 5V bulk: module has 2x22uF; 100uF added near the Pico/ports.
6. Triggers: TRIG_IN accepts 3.3–24 V (opto). TRIG_OUT is open collector with a jumper-selected pull-up. Pin orders: see each symbol. Main supply cable (2x4, pins 1–8 row-major): 1,8 VSUPPLY (12 V) | 2,7 VGND | 3 TRIG_IN | 6 TRIG_OUT | 4 SOFTSTOP+ | 5 SOFTSTOP-. SOFTSTOP+ -> 1k -> GP16 (10k pull-up to 3V3, 5.1 V zener to GND). SOFTSTOP- = GND: loop closed = run, broken = stop. Keyed shrouded IDC box headers (cannot be mated rotated) are used for this connector and ENC1–4. Measured system current was <= 1 A at 8.4 V, well inside ~3 A per 0.1in pin.
7. Encoders: 4 x 2x4 headers (pins 1–8: GND, CS, SCK, MOSI, MISO, 3V3, 5V, GND) on one SPI1 bus: CS = GP13/GP9/GP8/GP7. MT6835 uses 3V3 (3.3 V logic); RLS Orbis uses 5V with 3.3 V logic. Check MISO tri-states when CS is high. The four ENC headers are a generic SPI port (SPI1: SCK GP10, MOSI GP11, MISO GP12, own CS each, 3V3 and 5V supply), so any 3.3 V SPI device can be plugged in, not just encoders. Each device needs its own CS and an MISO that tri-states when CS is high.
8. Use shielded twisted cable for the gauge wires, shield to GND at the ADC end only.

HastaArduino
Sheet: /
File: hastadaq.kicad_sch
Title: HastaDAQ
Size: A2 | Date: | Rev: A
KiCad E.D.A. 10.0.6 | Id: 1/1